



USER GUIDE

Troubleshooting

Diagnose controller, camera, and game connections.

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Start at the first stage that fails: camera, recognition, Controller delivery, console reception, emulator selection, then the displayed game. Keep a normal gamepad available. Change one setting at a time so you know what fixed the issue.



Pixel Pal gives a friendly stop-and-check signal

The **VirtualGlove Controller (Arduino UNO Q)** hosts Setup, Glove Academy, and Help. Use **Help -> This console** for addresses specific to your installation. The examples below use placeholders, not addresses that every build shares.

Buster package source moved

An older RetroPie image may stop during installation with a message that [raspbian.raspberrypi.org/raspbian buster](https://raspbian.raspberrypi.org/raspbian/buster) has no Release file. Buster has moved to Raspbian's legacy archive. This is an operating-system package-source issue, not a VirtualGlove package failure. The installer detects it before changing the VirtualGlove payload.

A newly imaged, supported RetroPie system is the best long-term fix. To keep an existing Buster cabinet running, back up its source list and replace only the retired Raspbian host:

```
SH
sudo cp /etc/apt/sources.list /etc/apt/sources.list.before-virtualglove
sudo sed -i \
's|http://raspbian.raspberrypi.org|https://legacy.raspbian.org|g' \
/etc/apt/sources.list
sudo apt-get update
```

If the installer's message names a file under `/etc/apt/sources.list.d/`, back up and make the same host-only replacement in that file. **Do not replace** archive.raspberrypi.org/debian; it is a different Raspberry Pi repository. After `apt-get update` succeeds, rerun the same RetroPie installer. Existing pairing and settings remain intact.

The website will not open

1. Check power and give the Controller time to finish starting.
2. Put your browser device on the same reachable LAN. Check the router's client list for the Controller's current IP address.
3. Try <http://CONTROLLER-IP:8100/setup>. Secure pairing uses <https://CONTROLLER-IP:8443/setup>.
4. If the IP works but `.local` does not, investigate hostname resolution and guest-network/client isolation. With Wi-Fi and USB Ethernet connected, the Controller can have more than one address.

Use the plain `/setup` address; no `?ui=2` suffix is needed. Old query-string bookmarks still open Setup.

The first installer says the Controller name is already in use

Another device answered for the requested `.local` name. Rerun the installer and choose a distinct short name, such as `virtualglove-den`. Do not disconnect the other device merely to bypass the check: duplicate `.local` names can change automatically and break browser bookmarks or console delivery. Updates do not ask this question and never rename an installed Controller.

The secure page shows a privacy warning

On the first visit, compare the website certificate with the ID on the physical Controller matrix. When they match, use **Trust this Controller** on secure Setup to download and install this Controller's public authority certificate. Trust must be enabled once on each phone or computer. The private authority key is never downloaded.

If a previously trusted Controller starts warning after an ordinary upgrade, confirm that the browser is using the same hostname and inspect the certificate before proceeding. Do not casually reset trust. If the authority files are actually damaged, stop VirtualGlove, move `data/tls` to a specially named backup, and start VirtualGlove again. This deliberately creates a new Controller identity: compare its new Matrix ID and install its replacement authority on each browser device.

Do not change pairing keys to fix an unreachable website. If SSH works, use the [configuration troubleshooting reference](#) for application checks. The documented hardware may restart after Shutdown; a disappearing page alone does not prove that power has been removed.

Networking is red or grey

The fourth Setup marker and fourth Off-mode pixel represent a physical **Wi-Fi or Ethernet link**, including supported Ethernet adapters in USB docks.

Marker	Meaning	Next check
Green	At least one detected physical Wi-Fi/Ethernet link is up	Check console-service and authenticated-response markers next
Red	Detected relevant links report disconnected	Check wireless association, Ethernet cable, dock power, and upstream data connection
Grey	The host report is missing, stale, incomplete, or has no recognised interface	Check that the Controller host sampler was installed/upgraded; do not assume the cable is disconnected

Docker bridges and loopback do not make this marker green. A green link does not prove an IP address, Internet access, console reachability, or game delivery. Older Wi-Fi-only telemetry can confirm a connected wireless link but cannot rule out Ethernet when Wi-Fi is down. Use the current Controller installer or upgrade helper to update the host sampler.

The camera view is missing

Gestures off normally closes the camera. Open **Play** or **Glove Academy** and wait for **Starting camera and gesture tracking** to finish. First startup can take longer than switching between active profiles.

When no camera is connected, Dashboard deliberately shows **Camera unavailable** instead of repeatedly presenting a broken preview. This behaviour belongs to the Controller and is the same for RetroPie, Recalbox, Batocera, and LaunchBox; it is not a Recalbox-specific failure.

If no camera appears, check the powered USB hub, cable, and camera connection. On the Controller host, [lsusb](#) should show the camera. If it is absent there, the problem is below hand recognition. The documented helper attempts one controlled recovery for the enrolled camera. A hub that proves per-port power support cycles only the saved camera port; a guarded whole-hub fallback is used only when the hub carries no networking. The Controller does not report success until it reads a real frame. Repeated setting changes will not repair a disconnected USB device. See [Camera selection](#).

The camera works but the hand is not recognised

Keep one whole hand in frame with its palm facing the camera. Light the hand from the camera side and avoid a bright window behind it. Try a bare hand before changing glove settings; the glove-colour option is only a diagnostic label. Use the first Glove Academy lesson to verify basic detection.

If detection repeatedly disappears, fix visibility before tuning gesture thresholds. Tracking losses and ordinary stationary jitter are different problems and should be reported separately.

Current releases install MediaPipe Hands 0.10.35 as the single Controller runtime. There is no user-selectable old-runtime fallback. If the worker reports that its wheel is missing, rerun the current Controller installer rather than adding a runtime setting to [device.json](#).

The hand is detected but the game does not move

1. Close local Play and Glove Academy; they pause cabinet input. Finish tuning, then explicitly start controller delivery if required.
2. Check the selected player and any request to set a fresh centre. Selecting a player loads their saved centre automatically. Use **Centre hand** if no centre is saved or the camera or playing position has changed.
3. Select **Start controller**. Armed means delivery is permitted when a valid game session or intentional manual profile is active; it does not mean packets are always being sent.
4. Check Setup's console-service and authenticated-response markers. A reachable service with unconfirmed authentication suggests pairing needs attention. Neither marker proves emulator input consumption.
5. Confirm that the game has actually started in RetroArch. The exact ROM filename must be registered; [.nes](#), [.zip](#), and [.7z](#) are separate entries.
6. Check the emulator and controller selection. For native Super Glove Ball, choose Nestopia (VirtualGlove); for its joystick fallback choose FCEUmm.

If Controller Router is enabled, return the hand to neutral once after the game appears. Router deliberately rejects directions and button gestures until that fresh neutral observation, while physical controls remain available immediately. If no physical controller responds until VirtualGlove is stopped, the console has an older Router build; close the game and rerun the current console installer.

If the ROM was added after VirtualGlove was installed, refresh the frontend's game list before testing it. Recalbox and Batocera also need a VirtualGlove service restart or reboot after a newly registered Super Glove Ball ROM so the missing exact-ROM native choice can be created. In LaunchBox, rerun the current installer if an older installation still points **VirtualGlove RetroArch** at a batch file or leaves standard RetroArch assigned to an NES game. The current installer uses the Python bridge directly and migrates standard RetroArch NES assignments while preserving genuinely different emulator overrides. The bridge also ensures the managed receiver on every launch. If Dashboard shows **Controller connection stopped** while a LaunchBox game is active, close the game and LaunchBox, rerun the current installer, and relaunch the game. The upgrade stops duplicate receivers left by an older Python environment without changing ROMs, saves, pairing, or the game registry.

If native hand movement reaches Super Glove Ball on LaunchBox but the V-sign Start gesture or thumbs-up Select gesture does not, close RetroArch and rerun the current installer. Current builds keep native gestures on the guarded Power Glove channel instead of also sending ordinary RetroPad input. Dashboard recognition plus a working physical joypad does not by itself prove that an older Windows receiver has this correction.

On generic RetroPie, confirm the separate [VirtualGlove](#) input device and its Player 1 mapping. If optional Controller Router is enabled-or on Recalbox and Batocera-open its Setup card or run [virtualglove-controller-router check](#). Confirm each saved source is connected, every enabled **VirtualGlove Merged Player 1-4** output exists, and the current RetroArch player indexes are assigned. Merged devices are intentionally neutral in EmulationStation and become active for physical input during Libretro gameplay. VirtualGlove gesture input remains limited to supported FCEUmm, stock Nestopia, and native Super Glove Ball paths. On Recalbox, the installer check must report **Persistent Libretro routing override**. Its generated [retroarchcustom.cfg.overrides.cfg](#) is expected to be rewritten at launch; do not repair that temporary file manually. On LaunchBox, the installer must report [network-retropad](#), a random high loopback port, and a validated isolation rule. A reported key conflict affects only the real-keyboard backup; VirtualGlove and physical XInput remain available. If gestures are recognised but FCEUmm does not move, rerun the current installer to restore the managed ordinary-game configuration and receiver route rather than changing global RetroArch settings. If LaunchBox reports that Nestopia (VirtualGlove) is missing or changed, rerun the matching VirtualGlove Windows installer. The affected game continues in FCEUmm joystick mode. On Batocera, an **ACTION** result for the packaged native core means the architecture could not be resolved or the on-console load test failed; leave FCEUmm selected and do not copy a core from another target. Verify the matching platform service from the [Installation Guide](#) before editing RetroArch settings.

If a fresh Recalbox/Batocera update reports several possible initial Player 1 controllers, run the installer with [--list-player1-devices](#), identify the intended pad, and repeat it with [--player1-device DEVICE-ID](#). Two identical, non-serialized pads are not guessed. If the selected pad disconnects during play, only its held state releases; VirtualGlove remains available. Reconnect that saved pad, or explicitly select its replacement. Seeing no response from the merged device in EmulationStation is expected-it deliberately becomes active only in RetroArch.

A gamepad does not control the game

Try another connected gamepad. The one in your hands may be assigned to Player 2 while another pad is Player 1. In Setup > Controller Router, check which pad belongs to each player. Close the game before changing an assignment, then relaunch it to use the corrected routing.

A wireless controller falls asleep or wakes during a game

Router keeps the named merged outputs connected while physical controllers sleep, wake, disconnect, or reconnect. A source returns to its saved player. RetroArch 1.19.1 receives temporary routing settings resolved immediately before launch; supported newer builds use strict named-device reservations.

If Router itself restarts or loses its merged outputs, exit the game, wait for Router to become ready, then relaunch. The running game cannot safely recover its former slots after its virtual controllers have been destroyed. Changes made in Setup > Controller Router take effect on the next launch.

Controller Router reports an unavailable controller

Select **Check controllers**, then press a direction or button on every connected pad during the ten-second test. Each result includes the controller name, its stable six-character identity suffix, and its assigned player. For example, [Wireless Controller · cd8a4a · Player 1](#) identifies the saved device without depending on a changing Linux event number.

If a controller is listed as **Unavailable**:

1. Reconnect that exact controller and wait for EmulationStation to recognise it.
2. Confirm it still appears and works in EmulationStation.
3. Run **Check controllers** again and press one of its controls.
4. If the old controller was replaced, close every RetroArch game, assign the replacement in Controller Router, and save. Router never silently substitutes a different device.

A connected controller can report **Mapping refreshed** after a valid remap in EmulationStation. Its player assignment is retained and the refreshed mapping is used at the next game launch. If one controller responds while another is missing, Setup reports both facts rather than treating the whole test as a pass.

If no physical controller is assigned to Player 1, Setup shows a red warning. Gameplay may still work, but the platform's RetroArch menu and exit hotkeys may be unavailable. Assign at least one physical controller to Player 1 unless a keyboard is deliberately kept available.

Controller Router manages Libretro gameplay, not standalone emulator programs. Original controllers continue to navigate EmulationStation. Do not use the lack of merged-device movement in the frontend as evidence of a failure.

A filename such as [Gun.Smoke \(USA\).7z](#) must keep its punctuation in the registry even though the displayed game name is **Gun Smoke**. See [Register games](#) and the [Gameplay Guide](#).

Pairing asks for more than one code

Both pairing methods need the six-digit approval PIN displayed on the Controller matrix and the certificate-ID comparison. **Code pairing** additionally uses the one-time code generated on the selected console. **Password pairing** additionally uses that console's SSH username and password. The two codes are not interchangeable.

If confirmation expires, select **Start a new confirmation**. The saved console and method stay fixed during the two-minute window; change them after it ends. A failed submitted request also requires fresh confirmation. Save console edits with **Save settings** before pairing. The separate console one-time code remains valid for five minutes. Setup shows the correct command after RetroPie, Recalbox, Batocera, or LaunchBox is selected and saved. LaunchBox uses one-time-code pairing only. Run the displayed PowerShell command as the same Windows user who runs LaunchBox. If Windows asks about network access, allow the Python runtime on Private networks only. VirtualGlove's authenticated receiver runs in that signed-in desktop session, but ordinary game input reaches RetroArch through the managed loopback RetroPad and is not conditioned on window focus. A focus change must not release an active held control; tracking loss, controller timeout, game exit, or receiver shutdown does. If more than one console is online, run it on the exact console named in Setup; a code displayed by a different console cannot open the intended listener. A platform-mismatch error means the saved selection does not match the operating system detected by that console; correct and save the selection before retrying. When a submitted pairing attempt finishes, the matrix releases the approval PIN and resumes its normal display. When idle, the glove animation follows your On, Dim, or Off attract setting; active game and status displays still take priority. A completed attempt should not leave the old PIN scrolling for the rest of its two-minute window.

Use the [pairing walkthrough](#); never paste pairing tokens, passwords, or live approval PINs into a public support report.

If a saved DHCP address changes or `.local` is briefly unavailable, leave the pairing key in place. Both controller input and game-profile delivery can discover the paired peer on the same ordinary LAN and resume by authenticated unicast. Guest isolation, VLANs, or blocked local broadcasts can prevent discovery; enter a current address or repair local name resolution in that case rather than pairing repeatedly.

Movement drifts or feels reversed

Check the selected game profile and centre before changing sensitivity. Hold a relaxed hand at the intended playing position and choose **Centre hand**. Support your forearm where practical. Re-centre after moving the camera.

A profile such as Program D intentionally reverses controls. Native Super Glove Ball and joystick-style mappings behave differently, so verify the selected core. Programs 4, 9, 10, 13, and 14 also use deliberately specialized controls rather than the ordinary position-based D-pad. Before retuning recognition, compare the selected profile with the [complete Programs 1-14 gesture cards](#). If ordinary movement is correct but a gesture is unreliable, use **Glove Academy -> Tune gestures** and describe that symptom to Pixel Pal.

For FCEUmm digital directions, Setup's **Joystick dead zone** adjusts how far the selected player moves beyond the square centre box. Inside or on its boundary, all positional directions release; beyond a side is a cardinal direction and beyond a corner is a diagonal. Start with **Use standard size**, then save one small change at a time while watching the live direction indicators. The box is anchored to the hand centre saved by **Centre hand**, and its effective width and height are at least 1.5 times the saved hand size. Near an edge it moves inward intact. Live hand size and resting jitter do not make it change. Slider changes preview immediately; Save applies them to gameplay. This setting does not change native Super Glove Ball X/Y travel or cure processing latency. Use reach controls under **Glove Academy -> Tune gestures -> Movement reach** for native screen coverage and the latency procedure below for delay. Smaller reach values need less physical hand travel. **Latest coordinate** is the tested default; it is the only live native movement behaviour and clamps at the saved reach edges. Historical bounded-curve replay is an engineering tool, not a Dashboard setting. If the Robo-Glove still jumps after the hand leaves and re-enters the picture, confirm that the Controller and RetroPie are on the same current release before changing reach or smoothing. Version 0.4.2 retains the guard for one contradictory or unusually distant non-forward reacquisition for one fresh result while allowing strongly aligned forward movement immediately.

Start triggers accidentally, or a gesture stays active

Practise the V sign and its release in Glove Academy. Keep your fingers clearly away from a menu pose while performing another action. Use **A gesture happens accidentally** in Tune gestures if recognition needs personalisation.

Menu Guard suppresses D-pad and button output; native continuous positioning still follows the hand. Select **Stop controller** for a dependable pause while repositioning. Do not use extra smoothing to hide a recognition problem.

Controls feel delayed

Close optional camera previews during gameplay. Keep lighting and camera setup stable, then compare deliberate movements and supported stationary holds. Avoid changing several camera, core, and display settings at once.

If movement pauses and then catches up in a burst, check the connection before changing gesture or camera settings. This can happen on any supported platform when the Controller or console uses Wi-Fi: delayed packet delivery can look like slow or jerky recognition even when the hand is tracked correctly. Compare the same movement with a wired connection, if available. If the physical joypad is smooth but VirtualGlove catches up in bursts, investigate the Controller-to-console network path; if both lag, check the emulator and display path too. This is a diagnostic possibility on RetroPie, Recalbox, and LaunchBox, not a confirmed platform-specific fault or a reason to change their network settings by default.

Batocera has a confirmed Wi-Fi power-saving case. VirtualGlove turns off power saving on connected Wi-Fi interfaces when its service starts and again before a game, including the next launch after a wireless reconnection. Wired and disconnected Wi-Fi interfaces are unaffected. If movement still pauses and jumps, run `iw dev wlan0 get power_save` on the Batocera console (substitute the connected wireless interface if it is not `wlan0`); **Power save: off** is expected while VirtualGlove is running. Check the Controller's link as well: the Batocera setting cannot change power saving on the Controller or router.

Software status can locate processing delays but cannot measure the complete hand-to-screen delay. Follow the layered method in the [Engineering Journey](#) before drawing conclusions from screenshots or timestamps on different computers. Use the [Engineering Toolkit](#) only when the ordinary camera and connection checks do not identify the cause.

My player or backup looks wrong

Check **Active player** first: player selection applies across browsers. Progress and settings live on the Controller, not in browser storage. Each downloaded backup contains only the selected player's hand setup and excludes Academy progress. Your browser usually puts it in Downloads with a player-based name, such as `alex-virtualglove-hand-setup.json`.

Restore updates the selected player after review. An empty personal-threshold object can simply mean defaults are in use. Version-3 backups carry the centre-box size and effective gesture sensitivity; version-2 backups are migrated on import. See [backup locations and restore choices](#).

What to include when asking for help

Start with **Setup -> Download system report**. It records the relevant versions, camera/runtime choices, active profile/input mode, and connection-check results but omits video, secrets, personal hand data, ROM names, and network addresses.

Record the exact software commit and matrix firmware from the page footer, Controller board variant, camera/dock models, connection type, selected player and profile, core, game filename, and steps that reproduce the symptom. Say whether it occurs in Academy, local Rock Paper Scissors, or only in a cabinet game. Include the four status results and any tracking losses.

Share a short relevant error excerpt or aggregate diagnostic report. Exclude credentials and private video. Raw latency recordings should remain temporary and local unless you explicitly choose to share them. See [Contributing](#) for the project's testing and reporting workflow.